Persistent Demographic Information in X-ray Foundation Embeddings: a Risk for a Safe and Fair Deployment in Healthcare

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Abstract

Medical imaging vector embeddings (vembs) from foundation models improve efficiency in medical imaging tasks but may leak sensitive demographic information, raising concerns about fairness and privacy and limiting their safe deployment in real world. In this paper, we analyze chest X-ray embeddings from MIMIC-CXR and CheXpert using two foundation models (CXR Foundation, BiomedCLIP), revealing significant and persistent leakage across demographic information. We investigate whether demographic attributes—age, sex, ethnicity, and insurance type—are implicitly encoded in patient representations. Using predictive modeling, we show that machine learning models can reliably infer these attributes, even when they are not explicitly included. We also identify specific embedding dimensions associated with demographic prediction and demonstrate that altering or removing them has minimal impact on task performance. These findings expose persistent fairness vulnerabilities in medical AI systems and are indicative that demographic information is deeply and robustly encoded, making simple feature removal strategies inadequate for bias mitigation.

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1. Introduction

The integration of machine learning into healthcare has accelerated with large datasets like MIMIC-CXR (Goldberger et al., 2000) and CheXpert (Irvin et al., 2019), enabling foundational models that generate compact vector representations (embeddings) from medical images (Huang et al., 2023; Qiu et al., 2023). Vector embeddings (vembs) enhance downstream tasks by reducing computational costs and facilitating transfer learning (Huang et al., 2023). Despite their advantages, these embeddings raise significant ethical concerns. A key issue is that vembs may inadvertently encode sensitive demographic attributes—age, sex, ethnicity, and socioeconomic status—potentially introducing biases that propagate through diagnostic models. If unaddressed, these biases can distort predictions and lead to inequitable healthcare outcomes. For instance, models biased toward particular demographic groups may perform poorly for underrepresented populations, exacerbating health disparities (Wu et al., 2025; Seyyed-Kalantari et al., 2020; 2021; Glocker et al., 2023; Rajpurkar et al., 2022; Vaidya et al., 2024; Larrazabal et al., 2020; Chen et al., 2020a; Wiens et al., 2019; Vyas et al., 2020; Gichoya et al., 2022; **?**). This study investigates how extensively demographic information is encoded within medical image embeddings, focusing on contrastive models that serve as feature extractors for downstream prediction tasks (Chen et al., 2020b; Sellergren et al., 2022; Zhang et al., 2023). More in detail, we explore how well demographic information are redundantly encoded in the embedding space of the selected foundation models, offering actionable insights for building safer clinical AI systems.

We analyze visual embeddings produced by two foundation models (Sellergren et al., 2022; Zhang et al., 2023) trained on chest radiographs from the MIMIC-CXR and CheXpert datasets. To assess demographic information encoded in these representations, we train machine learning models to detect distributional shifts and recover demographic attributes. Our findings reveal how demographic signals persist in vembs, raising concerns about bias in AI-driven medical imaging and emphasizing the need for more equitable embedding methods in clinical applications.

2. Materials and Methods

In this study, we leverage two pre-existing datasets that required no additional ethical approval:

- MIMIC-CXR: Contains 228,905 radiographs collected between 2011-2016, annotated with 14 categories including the "No Finding" label (indicating the absence of any pathology or injury) and demographic information (age, sex, ethnicity, insurance type) (Goldberger et al., 2000).
- CheXpert: Comprises 224,316 radiographs from Stanford Hospital (1998-2017) with 14 categories including the "No Finding" label and demographic information (age, sex, ethnicity) (Irvin et al., 2019).

To enhance analytical robustness, we standardized ethnicity categories into five consolidated groups and excluded underrepresented demographic segments (<5% of dataset). Additionally, for CheXpert, we removed radiographs with mismatches between patient metadata and available images. This filtering yielded 200,452 and 198,836 radiographs from MIMIC-CXR and CheXpert, respectively, for comprehensive analysis.

2.1. Vector Embedding Extraction Systems

We extracted embeddings from chest X-rays using two state-of-the-art foundational models:

- CXR Foundation: Employs a three-phase approach (non-medical pretraining, supervised contrastive learning on chest X-rays, task-specific fine-tuning) generating 1376-dimensional embeddings (Sellergren et al., 2022; 2023).
- BiomedCLIP: Trained on the PMC-15M biomedical image-text corpus, creating compact 512-dimensional embeddings (Zhang et al., 2023)

2.2. Analytical Framework

Our investigation followed a dual methodological approach:

2.2.1. Predictive Modeling Assessment

To assess the extent to which demographic attributes are encoded and could retrieved from the embeddings, we trained Multi-Layer Perceptron (MLP) classifiers to predict each demographic attribute from the vembs. Stratified 10-fold cross-validation was applied with patient-level isolation to ensure no information leakage between training and test sets (Bradshaw et al., 2023). Model performance was evaluated using both F1-score and ROC-AUC, providing a robust measure of predictive accuracy and averaged across folds.

2.2.2. Feature Removal

To investigate which embedding features contribute most to demographic prediction, we computed input gradients with respect to the model output and ranked features by their average absolute gradient magnitude. We then perform iterative feature ablation, zeroing out the top-k% most informative features (for $k = 10, 20, 50$). At each step, the model's predictive performance was reevaluated after retraining. ROC-AUC was used as the primary evaluation metric, as it provides a consistent baseline of 0.5 regardless of class imbalance or the number of classes.

3. Results

3.1. Study Sample Characteristics

We analyzed 200,452 radiographs from MIMIC-CXR (mean age 62.5±16.5 years) from 46,187 patients and 198,836 radiographs from CheXpert (mean age 62.8±17.4 years) from 57,270 patients.

3.2. Demographic Prediction Performance

According to Table 1, MLP models achieved strong predictive performance for binary demographic attributes, particularly sex and the “No Finding” disease label. In both MIMIC-CXR and CheXpert datasets, models consistently yielded ROC-AUC scores exceeding 0.95 for sex prediction, indicating that this attribute is prominently encoded in the learned representations. Performance for age, ethnicity, and insurance type was lower but still above random chance. These results suggest that demographic information, especially sex, is readily recoverable — even when not explicitly encoded during training.

3.3. Feature Removal

Table 2 shows results for feature removal analysis. Perturbing the top-k % most influential embedding dimensions (k=10, 20, 50) had minimal impact on model performance across all demographic prediction tasks. For instance, sex prediction ROC-AUC dropped by less than 0.01 in all embedding types, indicating high robustness. Similarly, age, ethnicity, and insurance type predictions showed negligible changes in ROC-AUC after perturbation. These results suggest that demographic information is not confined to a small subset of embedding dimensions but is instead redundantly distributed across the entire embedding space.

4. Discussion

Our findings confirm that demographic information is extensively encoded across embedding features with significant redundancy and can be effectively retrieved using machine

Table 1. Performance metrics (F1-score and ROC-AUC) for MLP model trained on CXR-Foundation and BiomedCLIP embeddings

Dataset	**Model**	**Task**	**Metrics**	
			F1-score	**ROC-AUC**
MIMIC-CXR	**CXR Foundation**	Sex	95.59 ± 0.17	99.19 ± 0.05
		Age	67.19 ± 0.54	88.93 ± 0.21
		Ethnicity (Binary)	75.70 ± 0.48	84.48 ± 0.21
		Ethnicity (Multi-Class)	44.98 ± 1.78	79.85 ± 0.32
		Insurance	46.76 ± 3.06	76.22 ± 0.17
		Disease (No finding)	77.30 ± 0.40	86.41 ± 0.11
	BiomedCLIP	Sex	84.29 ± 0.07	92.57 ± 0.05
		Age	47.58 ± 0.40	77.77 ± 0.04
		Ethnicity (Binary)	62.32 ± 0.29	72.40 ± 0.07
		Ethnicity (Multi-Class)	30.43 ± 0.37	69.40 ± 0.39
		Insurance	39.35 ± 0.24	69.82 ± 0.07
		Disease (No finding)	73.64 ± 0.33	82.89 ± 0.03
CheXpert	**CXR Foundation**	Sex	94.29 ± 0.19	98.70 ± 0.06
		Age	67.15 ± 0.32	88.75 ± 0.16
		Ethnicity (Binary)	67.71 ± 0.48	74.29 ± 0.31
		Ethnicity (Multi-Class)	41.17 ± 1.43	74.57 ± 0.62
		Disease (No finding)	83.69 ± 0.36	95.52 ± 0.06
	BiomedCLIP	Sex	81.43 ± 0.18	90.06 ± 0.07
		Age	46.58 ± 0.44	75.89 ± 0.02
		Ethnicity (Binary)	60.22 ± 0.52	66.33 ± 0.07
		Ethnicity (Multi-Class)	27.77 ± 0.79	66.52 ± 0.27
		Disease (No finding)	75.73 ± 0.58	91.74 ± 0.03

learning models. This extends previous research showing demographic encoding in reduced-dimensional embeddings to multiple embedding types and demographic attributes (Glocker et al., 2023). It is important to notice that such hidden sensitive information highlights practical concerns around clinical fairness, as learned correlations may be exploited as shortcuts, introducing systemic biases (Burns et al., 2023; Gichoya et al., 2022; Jabbour et al., 2020). These risks are enhanced by the apparent redundancy of demographic encoding, which complicates debiasing strategies (Watson et al., 2022; Salvado et al., 2024; Konate et al., 2025).

In real-world clinical contexts, such leakage may result in models implicitly using demographic proxies like ethnicity or insurance status, reinforcing healthcare disparities. Our analysis reveals how these information is spread across embeddings, offering insight into where and how interventions might be targeted. These results underscore the importance of informed mitigation strategies such as adversarial debiasing (Zhang et al., 2018; Correa et al., 2024) or disentangled representation learning (Creager et al., 2019), especially for deployment in high-stakes environments.

5. Conclusion

This study demonstrates that vector embeddings derived from chest radiographs substantially encode demographic information. Rather than eliminating this information, which could obscure valuable clinical correlations, future work should focus on understanding and managing it effectively. Developing methods to disentangle demographic information from disease-specific features within embeddings could minimize bias risk in predictive models while preserving clinically relevant information. Additionally, techniques to audit and interpret embeddings could provide greater transparency in how these models utilize encoded information.

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Table 2. % feature perturbation (demographic prediction VS. disease prediction - mean performance across folds ± standard deviation).

Dataset	Model	Task	Percentage of features modified					
			0%	1%	5%	10%	20%	50%
MIMIC-CXR	CXR-F	Sex	99.2 ± 0.1	99.2 ± 0.0	99.1 ± 0.0	98.9 ± 0.1	98.7 ± 0.0	97.9 ± 0.1
		Disease*	86.4 ± 0.1	86.4 ± 0.1	86.3 ± 0.1	86.3 ± 0.1	86.4 ± 0.1	86.3 ± 0.1
		Age	88.9 ± 0.2	88.7 ± 0.1	88.6 ± 0.2	88.2 ± 0.1	88.0 ± 0.1	87.4 ± 0.1
		Disease*	86.4 ± 0.1	86.4 ± 0.1	86.3 ± 0.2	86.4 ± 0.1	86.3 ± 0.1	86.4 ± 0.1
		Eth. B.	84.5 ± 0.2	84.2 ± 0.3	83.9 ± 0.2	83.3 ± 0.2	82.6 ± 0.2	80.9 ± 0.2
		Disease*	86.4 ± 0.1	86.3 ± 0.2	86.3 ± 0.2	86.3 ± 0.1	86.3 ± 0.2	86.3 ± 0.1
		Eth. M.	79.9 ± 0.3	79.4 ± 0.3	79.1 ± 0.2	78.7 ± 0.3	78.1 ± 0.2	76.4 ± 0.2
		Disease*	86.4 ± 0.1	86.4 ± 0.1	86.3 ± 0.1	86.3 ± 0.1	86.4 ± 0.1	86.3 ± 0.1
		Insurance	76.2 ± 0.2	75.8 ± 0.2	75.8 ± 0.3	75.8 ± 0.2	75.6 ± 0.2	75.4 ± 0.2
		Disease*	86.4 ± 0.1	86.3 ± 0.1	86.4 ± 0.1	86.4 ± 0.1	86.4 ± 0.1	86.3 ± 0.1
	B-CLIP	Sex	92.6 ± 0.1	92.2 ± 0.0	92.2 ± 0.1	92.1 ± 0.1	92.0 ± 0.1	91.7 ± 0.1
		Disease*	82.9 ± 0.0	82.8 ± 0.1	82.8 ± 0.1	82.8 ± 0.0	82.8 ± 0.0	82.8 ± 0.1
		Age	77.8 ± 0.1	77.0 ± 0.1	77.0 ± 0.1	76.9 ± 0.1	76.8 ± 0.2	76.7 ± 0.1
		Disease*	82.9 ± 0.0	82.8 ± 0.0	82.7 ± 0.0	82.7 ± 0.1	82.7 ± 0.1	82.8 ± 0.1
		Eth. B.	72.4 ± 0.1	72.1 ± 0.3	71.9 ± 0.3	72.0 ± 0.2	71.8 ± 0.3	71.8 ± 0.2
		Disease*	82.9 ± 0.0	82.7 ± 0.1	82.8 ± 0.0	82.8 ± 0.0	82.8 ± 0.1	82.8 ± 0.1
		Eth. M.	69.4 ± 0.4	68.2 ± 0.3	68.1 ± 0.2	68.1 ± 0.4	68.2 ± 0.2	67.8 ± 0.2
		Disease*	82.9 ± 0.0	82.7 ± 0.1	82.7 ± 0.1	82.8 ± 0.1	82.8 ± 0.1	82.7 ± 0.1
		Insurance	69.8 ± 0.1	69.5 ± 0.1	69.5 ± 0.1	69.4 ± 0.2	69.4 ± 0.1	69.1 ± 0.1
		Disease*	82.9 ± 0.0	82.8 ± 0.0	82.8 ± 0.0	82.8 ± 0.0	82.7 ± 0.1	82.8 ± 0.1
CheXpert	CXR-F	Sex	98.7 ± 0.1	98.7 ± 0.1	98.5 ± 0.0	98.3 ± 0.1	98.1 ± 0.0	97.1 ± 0.1
		Disease*	95.5 ± 0.1	95.5 ± 0.0	95.5 ± 0.1	95.5 ± 0.0	95.5 ± 0.0	95.4 ± 0.0
		Age	88.8 ± 0.2	88.5 ± 0.1	88.3 ± 0.1	87.9 ± 0.1	87.7 ± 0.1	86.8 ± 0.1
		Disease*	95.5 ± 0.1	95.5 ± 0.0	95.5 ± 0.0	95.5 ± 0.0	95.5 ± 0.1	95.5 ± 0.0
		Eth. B.	74.3 ± 0.3	74.4 ± 0.3	74.0 ± 0.3	73.3 ± 0.4	72.8 ± 0.1	71.4 ± 0.3
		Disease*	95.5 ± 0.1	95.5 ± 0.0	95.5 ± 0.0	95.5 ± 0.0	95.5 ± 0.0	95.4 ± 0.0
		Eth. M.	74.6 ± 0.7	74.3 ± 0.2	74.3 ± 0.3	73.8 ± 0.3	73.4 ± 0.4	71.8 ± 0.3
		Disease*	95.5 ± 0.1	95.5 ± 0.1	95.5 ± 0.0	95.5 ± 0.0	95.5 ± 0.0	95.4 ± 0.0
	B-CLIP	Sex	90.1 ± 0.1	89.7 ± 0.2	89.6 ± 0.1	89.5 ± 0.1	89.2 ± 0.1	88.9 ± 0.1
		Disease*	91.7 ± 0.0	91.6 ± 0.1	91.7 ± 0.0	91.6 ± 0.0	91.7 ± 0.0	91.7 ± 0.0
		Age	75.9 ± 0.0	75.1 ± 0.2	75.0 ± 0.1	74.9 ± 0.1	74.9 ± 0.2	74.8 ± 0.2
		Disease*	91.7 ± 0.0	91.6 ± 0.0	91.6 ± 0.1	91.6 ± 0.0	91.6 ± 0.0	91.6 ± 0.1
		Eth. B.	66.3 ± 0.1	65.8 ± 0.2	65.9 ± 0.2	65.8 ± 0.2	65.7 ± 0.2	65.4 ± 0.3
		Disease*	91.7 ± 0.0	91.6 ± 0.0	91.6 ± 0.0	91.6 ± 0.1	91.6 ± 0.1	91.6 ± 0.1
		Eth. M.	66.5 ± 0.3	64.6 ± 0.5	64.7 ± 0.3	64.7 ± 0.2	64.7 ± 0.3	64.8 ± 0.5
		Disease*	91.7 ± 0.0	91.6 ± 0.0	91.6 ± 0.1	91.7 ± 0.0	91.6 ± 0.0	91.5 ± 0.1

CXR-F – CXR Foundation, B-CLIP – BiomedCLIP, Eth. – Ethnicity, B. – Binary, M. – Multi-class
* Modifying key features to predict the demographic variable in the same group of tests

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